Jio Platforms Limited

DEPLOYMENT GUIDE

Consent Management System

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Version History

Date	Contributor	Contribution	Version	State
Nov 25, 2025	Kuldeep Prajapati	Initial Draft	0.1	Draft
Nov 26, 2025	Lokesh Sharma	Reviewed and released the document	1.0	Released

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Table of Contents

Version History *1*

1. Overview *3*

2. Open-Source Setup & Credential Management *4*

2.1. Open-Source Component Deployment 4

2.2. Credential Preparation 4

2.3. Environment Variables 4

2.4. Kubernetes Setup 5

2.5. MongoDB User Setup 5

2.6. Kubernetes Secrets Creation 5

3. Backend & Frontend Deployment *6*

3.1. Microservices Preparation 6

3.2. Backend Deployment 6

3.3. Frontend Deployment 6

3.4. Full Deployment Script 6

4. Verification Steps *7*

5. Troubleshooting / Common Errors *8*

6. Rollback / Cleanup *9*

7. Architecture Overview *10*

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1. Overview

This guide provides **step-by-step instructions** for deploying the Jio Consent Management System (JCMS). It is designed for both technical and non-technical users.

Technology Stack: - **Backend:** Spring Boot (Java 21) - **Frontend:** ReactJS, Node.js 20 - **Other:** Docker, Kubernetes (with ConfigMaps), MongoDB v7.0.14, Nginx - **Open-Source Components:** WSO2 v4.0.0, HashiCorp Vault v1.19.4, Nginx with Lua module, Apache Kafka (open source)

Purpose: Secure, compliant, and scalable consent and grievance management for Data Fiduciaries and Data Principals.

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2. Open-Source Setup & Credential Management

2.1. Open-Source Component Deployment

- **WSO2 Setup:**
  - Navigate to `JCMS/OpenSourceConfiguration/Wso2/`
  - Follow the README guide
  - Note down service URLs and ports
- **HashiCorp Vault Setup:**
  - Navigate to `JCMS/OpenSourceConfiguration/HashiCorpVault/`
  - Deploy Vault and configure secrets storage
  - Store private certificates, AES keys, and service credentials
  - Verify Vault URLs and token accessibility
- **Nginx Deployment:**
  - Deploy Nginx server with Lua module
  - Configure Lua scripts for consent and grievance endpoints

2.2. Credential Preparation

Credential Type	Purpose	Location / Notes
MongoDB Standard User	Backend DB access	`mongodb-secret`
MongoDB Admin User	Admin operations	`admin-mongodb-secret`
WSO2 Credentials	API access	Generated via WSO2 Cred Generator
Vault Token / Keys	Certificates & encryption	Accessible via Vault service
JWT / Auth Secrets	User & business authentication	Configured per microservice
Bhashini Translation Credentials	Multi-language support	Configured per translator API
DigiLocker Credentials	DigiLocker API integration	Stored securely in backend services

2.3. Environment Variables

```
NAMESPACE=<your_k8s_namespace>
SERVICE_ACCOUNT=<k8s_service_account>
MONGO_USER=<mongodb_user>
MONGO_USER_PASSWORD=<mongodb_user_password>
MONGO_ADMIN_USER=<mongodb_admin_user>
MONGO_ADMIN_PASSWORD=<mongodb_admin_password>
```

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```
MONGO_HOSTS=<host1,host2,...>
REPLICA_SET=<replica_set_name>
```

2.4. Kubernetes Setup

```
kubectl create namespace $NAMESPACE
kubectl create serviceaccount $SERVICE_ACCOUNT -n $NAMESPACE
kubectl create clusterrolebinding ${NAMESPACE}-binding --clusterrole=edit
--serviceaccount=${NAMESPACE}:${SERVICE_ACCOUNT}
kubectl create token $SERVICE_ACCOUNT -n $NAMESPACE --duration=8760h
kubectl config set-context --current --namespace=$NAMESPACE
```

- Configure kubeconfig for worker nodes using token and cluster info

2.5. MongoDB User Setup

Standard User:

```
db.createUser({
  user: "<MONGO_USER>",
  pwd: "<MONGO_USER_PASSWORD>",
  roles: [
    { role: "readWriteAnyDatabase", db: "admin" },
    { role: "dbAdmin", db: "cms_db_admin" }
  ]
});
```

Admin User:

```
db.createUser({
  user: "<MONGO_ADMIN_USER>",
  pwd: "<MONGO_ADMIN_PASSWORD>",
  roles: [{ role: "root", db: "admin" }]
});
```

2.6. Kubernetes Secrets Creation

```
kubectl create secret generic mongodb-secret -n $NAMESPACE --from-literal=
uri="mongodb://$MONGO_USER:$MONGO_USER_PASSWORD@$MONGO_HOSTS/?replicaSet=$
REPLICA_SET&authSource=admin"

kubectl create secret generic admin-mongodb-secret -n $NAMESPACE --from-li
teral=uri="mongodb://$MONGO_ADMIN_USER:$MONGO_ADMIN_PASSWORD@$MONGO_HOSTS/
?replicaSet=$REPLICA_SET&authSource=admin"
```

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3. Backend & Frontend Deployment

3.1. Microservices Preparation

- Read each backend microservice README
- Update ConfigMaps for ports, DB URLs, and dependent service URLs
- Verify all secrets exist
- Ensure logs hostPath directories exist or allow Kubernetes to create

3.2. Backend Deployment

- **Linux:** `./deployBackend.sh`
- **Windows:** `.\deployBackend.ps1`
- Verification:

```
kubectl get pods -n $NAMESPACE
kubectl get svc -n $NAMESPACE
kubectl logs <pod_name> -n $NAMESPACE
```

3.3. Frontend Deployment

- Update `.env` files
- **Linux:** `./deployFrontend.sh`
- **Windows:** `.\deployFrontend.ps1`
- Deploy after backend is up

3.4. Full Deployment Script

- **Linux:** `./DeployAll.sh` (if exists)
- **Windows:** `.\DeployAll.ps1` (if exists)
- Estimated time: 30–40 minutes

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4. Verification Steps

- Check pods: `kubectl get pods -n $NAMESPACE`
- Check services: `kubectl get svc -n $NAMESPACE`
- Access frontend portals:
  - Partner Portal
  - User Portal
  - Consent Popup

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5. Troubleshooting / Common Errors

Error	Cause / Solution
kubectl: command not found	Install kubectl
MongoDB authentication failed	Verify ConfigMap and secret values
Pod crash	`kubectl logs <pod_name> -n $NAMESPACE`
Service not reachable	Check ingress/service status
Deployment issues	Verify ConfigMap values

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6. Rollback / Cleanup

- Rollback if deployment fails or services are unstable
- Command: `kubectl delete namespace $NAMESPACE`

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7. Architecture Overview

- **Frontend:** Partner Portal, User Portal, Consent Popup
- **Backend:** Auth, Consent, Grievance, Audit, Notification, Cookies, Vault, Translator
- **Database:** MongoDB Replica Set
- **Messaging:** Apache Kafka (open source)
- **Security:** Vault
- **Open-Source Services:** WSO2, HashiCorp Vault, Nginx with Lua module

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